

User-Centered Design Principles for Decentralized Identity in Banking KYC

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Abstract

Decentralized identity (DID) systems promise improved privacy, user control, and interoperability in regulated digital identity verification, including in banking contexts. However, existing DID implementations often introduce usability barriers that limit adoption, particularly for non-technical users navigating Know-Your-Customer (KYC) onboarding—a mandatory process requiring banks to verify customer identity to prevent fraud and financial crime. This poster presents a UX-centred, four-phase mixed-methods investigation comparing a high-fidelity DID-based onboarding prototype with a traditional Web2 (centralised, platform-owned) KYC flow. Results show that DID significantly reduced task completion time (178s vs. 665s), improved perceived usability (SUS: 93.4 vs. 71.3), and increased user trust (6.1 vs. 5.1 on a 7-point scale). At the same time, wallet setup, private-key management, and the absence of onboarding-phase recovery guidance remain critical sources of user anxiety. We contribute five user-centered design principles for DID-enabled financial onboarding and discuss implications for designers working at the intersection of Web3 (decentralised, blockchain-based systems), fintech, and regulated digital services.

CCS Concepts

• **Human-centered computing** → **User interface design**; • **Security and privacy** → *Authentication*; • **Applied computing** → *Online banking*.

Keywords

Decentralized identity, KYC, user experience, Web3 usability, trust, onboarding design

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1 Introduction

Digital banking onboarding remains a high-friction experience [3]. Users must repeatedly re-enter personal data, upload identity documents multiple times, and accept opaque data-handling practices with limited transparency or control. Although Know-Your-Customer (KYC) processes are necessary for regulatory compliance, they erode trust and contribute to onboarding abandonment.

Decentralized identity (DID) systems propose an alternative in which users hold verifiable credentials in a digital wallet and selectively disclose identity attributes to service providers [1]. In practice, however, many DID implementations rely on unfamiliar interaction patterns, technical jargon, and abstract security concepts that are difficult for non-technical users to navigate [2]. These challenges are especially acute in banking, where usability failures can lead to exclusion from essential financial services.

Despite growing interest in DID, designers lack concrete, empirically grounded guidance on how to translate these concepts into usable, trustworthy financial onboarding. This work addresses that gap directly.

2 Study Overview

We conducted a four-phase mixed-methods study investigating usability, trust, and mental-model formation in DID-enabled banking onboarding. Two high-fidelity, debranded prototypes were developed in Figma and made interactive using ProtoPie: a Web2 KYC flow modelled on Coinbase's onboarding architecture (representing centralised identity verification via email, password, photo ID upload, and selfie), and a DID-based flow inspired by existing DID wallet interfaces (Privado ID and Dock Wallet), using wallet sign-in, verifiable credential review, and selective attribute disclosure. Neither prototype represents a live product.

2.1 Phase 1: Heuristic Evaluation

Two evaluators applied five of Nielsen's usability heuristics to the Web2 KYC prototype, identifying recurring friction points around transparency, repetitive data entry, missing progress indicators, and limited user control.

2.2 Phase 2: Interface Pattern Analysis

A structured review of two live DID wallet applications—Privado ID and Dock Wallet—was conducted against Bobrova & Perego's UCD guidelines [2], focusing on onboarding, credential management, and consent flows. Findings informed the DID prototype design and identified interaction patterns around wallet connection and recovery that shaped the subsequent co-design activities.

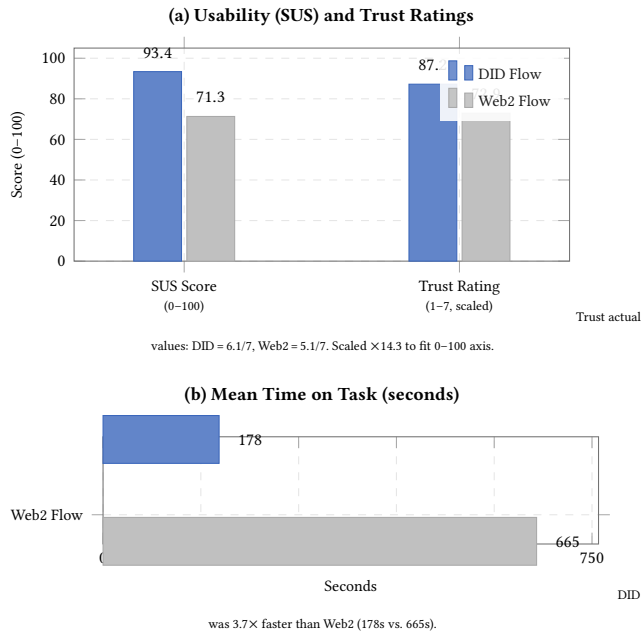


Figure 1: Quantitative outcomes ($n = 8$). (a) SUS usability score and trust rating, both on a 0–100 scale for comparability. (b) Mean time on task in seconds.

2.3 Phase 3: Co-Design Workshop

A remote participatory workshop ($n = 4$ non-technical users; students, designers, and professionals) combined journey mapping with group discussion. Sessions were conducted via Microsoft Teams and Miro, lasted 90 minutes, and were analysed thematically using Braun’s approach, mapped to Bobrova & Perego’s guidelines.

2.4 Phase 4: Controlled Usability Testing

Online, moderated, within-subjects, counterbalanced usability testing ($n = 8$; ages 18–44; 3F, 5M) compared both prototype flows. Participants completed onboarding tasks (sign-up, credential sharing, identity verification) and filled in post-task measures: SUS scores [5], 7-point trust Likert ratings, and NASA-TLX workload subscales. Sessions were screen-recorded and followed by semi-structured interviews analysed thematically.

3 Key Findings

Figure 1 visualises quantitative outcomes; Table 1 reports descriptive statistics. Qualitative themes are described below.

3.1 Dramatically Reduced Onboarding Time

DID onboarding was nearly four times faster than Web2 (178s vs. 665s), driven by elimination of repetitive manual steps such as document upload and multi-stage verification. This aligns with Nielsen’s efficiency heuristic [2].

Table 1: Usability testing outcomes: DID vs. Web2 ($n = 8$).

Metric	DID Flow	Web2 Flow
Mean SUS Score (0–100)	93.4 (SD 10.7)	71.3 (SD 19.6)
Task Completion (unassisted)	87.5%	62.5%
Mean Trust Rating (1–7)	6.1 (SD 1.4)	5.1 (SD 1.0)
Mean Time on Task	178s (SD 112)	665s (SD 451)
Mean Errors per Participant	0.4	2.7
NASA-TLX Mean (effort)	1.5	3.0

3.2 Higher Perceived Usability

DID achieved an “Excellent” SUS rating (93.4) against Web2’s “Good” (71.3) [5], confirming that DID interfaces can be made more intuitive than traditional KYC flows for non-technical users.

3.3 Increased Trust through Visible Consent

DID trust ratings ($M = 6.1$) exceeded Web2 ($M = 5.1$). Participants valued the explicit consent moment and credential visibility—the ability to see precisely which attributes would be shared before approving. Notably, some participants still associated Web2’s longer flow with a perception of thoroughness, suggesting trust is partly a psychological artefact of procedural complexity.

3.4 Stronger Sense of User Control

Users appreciated approving credential sharing rather than passively submitting documents. Post-task interviews surfaced a clear qualitative contrast between the two conditions:

“Web2 felt like gatekeeping, DID felt collaborative.” — P007, usability testing

This perception of collaboration—rooted in visible consent controls and the ability to inspect credentials before sharing—was the most consistent qualitative theme across both co-design and usability sessions. It suggests that trust in identity systems is not simply a function of technical security, but of how much agency the interface affords the user.

3.5 Fragile Mental Models around Wallet Concepts

Despite performance improvements, most participants struggled with wallet setup, private-key management, and the long-term implications of credential storage. Confusion was highest at the wallet-connection step, driving the 12.5% assisted completion rate for the DID condition.

3.6 Progressive Disclosure Mitigates Confusion

Co-design and usability data both showed that layered, contextual explanations—introduced at the moment of relevance rather than upfront—were consistently more effective at building comprehension without increasing cognitive load.

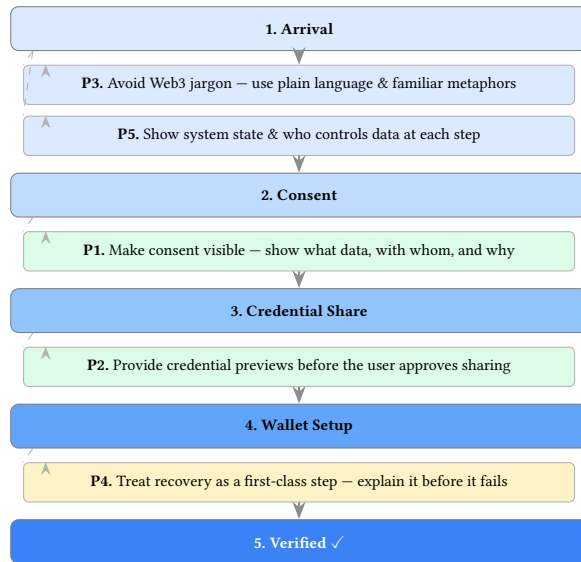


Figure 2: Five design principles mapped to the DID-enabled KYC onboarding journey.

4 User-Centered Design Principles

Figure 2 maps our five principles onto the DID onboarding journey. They are grounded in our empirical findings and extend Bobrova & Perego’s Web3 UCD guidelines into a regulated financial context [2].

4.1 P1. Make Consent Visible and Understandable

Interfaces should clearly communicate what data is shared, with whom, and for what purpose—at the moment of sharing, not buried in terms and conditions.

4.2 P2. Provide Credential Previews Before Sharing

Users should inspect the specific attributes within a credential before approving disclosure, supporting informed decision-making and reducing anxiety.

4.3 P3. Avoid Web3-Specific Jargon

Technical terminology (e.g., *private key*, *verifiable credential*) should be replaced with everyday language and familiar metaphors (e.g., “digital document”), introduced progressively at relevant moments.

4.4 P4. Treat Recovery as a First-Class Experience

Recovery options, their limitations, and the consequences of credential loss must be communicated *before* onboarding completes—not surfaced only after failure. This was the single most anxiety-inducing aspect of DID for participants [4].

4.5 P5. Communicate System State and Responsibility

Interfaces should explicitly show who controls identity data at each step, making the shift in custody from institution to user legible and non-threatening through step indicators, upfront document checklists, and empathetic microcopy.

5 Implications and Future Work

This study demonstrates that DID onboarding can meaningfully outperform traditional KYC across efficiency, usability, and trust when supported by intentional UX design. The findings challenge a common assumption: participants did not equate longer, more complex flows with greater trustworthiness—rather, transparency and control drove perceived trust. This shifts responsibility from backend security to interface design.

Three limitations bound this work. First, the study focused exclusively on the onboarding phase; the long-term UX of credential management, renewal, and revocation remains unexplored. Second, the small sample ($n = 8$) limits statistical generalisation. Third, expert review feedback was not obtained, representing a gap in practitioner validation.

Future work will test these principles through higher-fidelity deployments with compliance teams, explore applicability across government and healthcare identity contexts, and investigate how DID design translates across regulatory jurisdictions and lower digital-literacy populations.

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